ESP32 SMART DOOR LOCK

IoT PROJECT • SMART SECURITY • FACIAL RECOGNITION

PROJECT SUMMARY

Smart access control using ESP32

Developed an ESP32-based smart door lock that combines facial recognition, keypad authentication, LCD status display, and connected notifications for secure and automated door access.

01 **FACIAL RECOGNITION**

Automated user authentication for door access.

02 **KEYPAD + LCD**

PIN input and real-time access status display.

03 **TELEGRAM / IoT**

Connected notifications for security events.

ESP32 • CAMERA • FACE RECOGNITION • KEYPAD • LCD • TELEGRAM

SYSTEM IMPLEMENTATION

HOW THE SMART DOOR LOCK WORKS

01	FACE / PIN INPUT	User is authenticated through facial recognition or keypad PIN.
02	VERIFICATION	The controller checks whether the submitted identity or PIN is valid.
03	DOOR ACCESS	Valid authentication triggers the smart lock mechanism.
04	STATUS + ALERT	LCD shows the access status and connected notification can be sent.

KEY TAKEAWAYS

Secure authentication • Automated access • Real-time status • Connected monitoring